

HDOC Day 13 - Q1: Count Even and Odd Digits

```
#include <stdio.h>

int main() {
    int num, digit;
    int even = 0, odd = 0;

    printf("Enter a number: ");
    scanf("%d", &num);

    while (num != 0) {
        digit = num % 10;

        if (digit % 2 == 0)
            even++;
        else
            odd++;

        num = num / 10;
    }

    printf("Even digits = %d\n", even);
    printf("Odd digits = %d\n", odd);

    return 0;
}
```

Expected result: Example: Number = 12045 -> Even digits = 3, Odd digits = 2

HDOC Day 13 - Q2: Sum of Even and Odd Digits Separately

```
#include <stdio.h>

int main() {
    int num, digit;
    int sum_even = 0, sum_odd = 0;

    printf("Enter a number: ");
    scanf("%d", &num);

    while (num != 0) {
        digit = num % 10;

        if (digit % 2 == 0)
            sum_even = sum_even + digit;
        else
            sum_odd = sum_odd + digit;

        num = num / 10;
    }

    printf("Sum_even = %d\n", sum_even);
    printf("Sum_odd = %d\n", sum_odd);

    return 0;
}
```

Expected result: Example: Number = 12345 -> Sum_even = 6, Sum_odd = 9

HDOC Day 14 - Q1: Check Whether All Digits Are Unique

```
#include <stdio.h>

int main() {
    int num, digit, freq[10] = {0};
    int unique = 1;

    printf("Enter a number: ");
    scanf("%d", &num);

    while (num != 0) {
        digit = num % 10;
        freq[digit]++;

        if (freq[digit] > 1) {
            unique = 0;
            break;
        }

        num = num / 10;
    }

    if (unique == 1)
        printf("Unique\n");
    else
        printf("Not Unique\n");

    return 0;
}
```

Expected result: Examples: 12345 -> Unique; 12234 -> Not Unique

HDOC Day 14 - Q2: Frequency of Each Digit from 0 to 9

```
#include <stdio.h>

int main() {
    int num, digit, i;
    int freq[10] = {0};

    printf("Enter a number: ");
    scanf("%d", &num);

    while (num != 0) {
        digit = num % 10;
        freq[digit]++;
        num = num / 10;
    }

    for (i = 0; i <= 9; i++)
        printf("freq_%d = %d\n", i, freq[i]);

    return 0;
}
```

Expected result: Example: Number = 112333 -> freq_0=0, freq_1=2, freq_2=1, freq_3=3; all other frequencies are 0.